

Program -8

Create Kubernetes Pods using YAML Descriptors

1. Project Structure

```
Prgm8/  
|  
|---nodejs-configmap.yaml  
|---nodejs-pod.yaml  
|---server.js
```

2. Configuration files

Create **nginx-configmap.yaml**

```
apiVersion: v1  
kind: ConfigMap  
metadata:  
  name: nodejs-app-config  
  
data:  
  server.js: |  
    const http = require("http");  
    const port = 3000;  
    const server = http.createServer((req, res)  
=> {  
      res.end("Hello from Node.js running inside  
Kubernetes!");  
    });  
  
    server.listen(port, () => {  
      console.log(`Server running at port  
${port}`);  
    });
```

Create **nodejs-pod.yaml**

```
apiVersion: v1
kind: Pod
metadata:
  name: program-8
  labels:
    app: nodejs

spec:
  containers:
    - name: nodejs
      image: node:18
      command: ["node",
"/usr/src/app/server.js"]
      volumeMounts:
        - name: app-code
          mountPath: /usr/src/app
      ports:
        - containerPort: 3000

  volumes:
    - name: app-code
      configMap:
        name: nodejs-app-config
```

Create [server.js](#)

```
const http = require("http");  
const port = 3000;  
  
const server = http.createServer((req, res) =>{  
  res.send("hello from nodejs inside kubernetes");  
});  
  
server.listen(port, () =>{  
  console.log("server is running as port ${port}")  
});
```

3. Apply the yaml files

```
docker apply -f nodejs-configmap.yaml  
docker apply -f nodejs-pod.yaml
```

4. Get the pods

```
docker get pods
```

5. Access the nodejs application

```
kubectl port-forward pod/program-8 3000:3000
```

6. Check status

```
docker get pods  
docker get deployments
```

7. Expose the deployment as service

```
kubectl expose deployment nginx-deployment - --type=NodePort - --port=80
```

8. Access on the browser

```
http://localhost:3000
```

Output:-

```
fault
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl apply -f nodejs
-configmap.yaml
configmap/nodejs-app-config created
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl apply -f nodejs
-pod.yaml
pod/program-8 created
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubla     1/1     Running   1 (3n2s ago)  24h
program-8  0/1     ContainerCreating   0          27s
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl port-forward po
d/nodejs-pod 3000:3000
Error from server (NotFound): pods "nodejs-pod" not found
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl port-forward po
d/program-8 3000:3000
error: unable to forward port because pod is not running. Current status=Pending
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubla     1/1     Running   1 (4n7s ago)  24h
program-8  0/1     ContainerCreating   0          92s
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl get pods
NAME      READY   STATUS    RESTARTS   AGE
kubla     1/1     Running   1 (8m23s ago)  24h
program-8  1/1     Running   0          5m48s
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl port-forward po
d/nodejs-pod 3000:3000
Error from server (NotFound): pods "nodejs-pod" not found
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl port-forward po
d/nodejs-pod 3000:3000
Error from server (NotFound): pods "nodejs-pod" not found
1rv24mc082_rajatraj@rajatraj-Vostro-5581:~/devops/labprogram/prgn8$ kubectl port-forward po
d/program-8 3000:3000
Forwarding from 127.0.0.1:3000 -> 3000
Forwarding from [::1]:3000 -> 3000
Handling connection for 3000
Handling connection for 3000
[
```

Hello from Node.js running inside Kubernetes!